

A. Appendix

15

A.1 Supplementary Figures

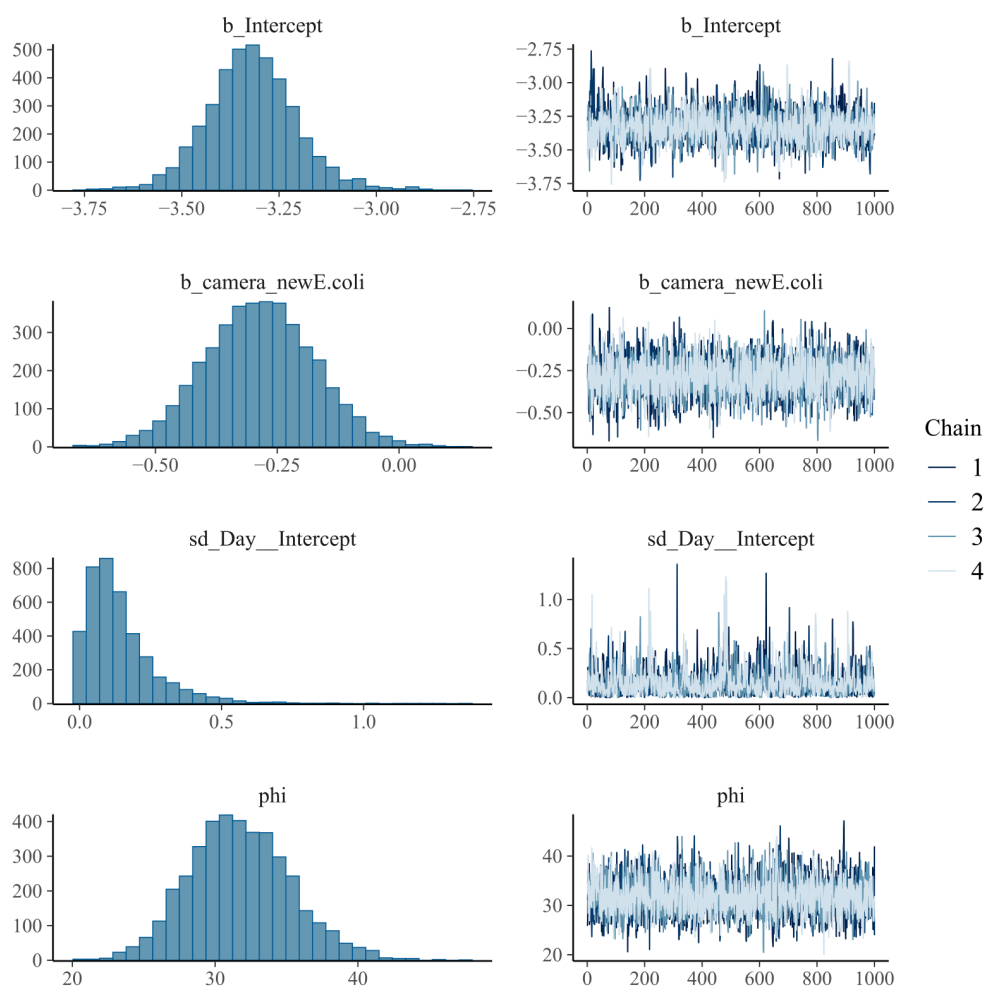


Figure S1. *brms* output for distance travelled, showing density plots for each factor (left) and caterpillar plots (right). Indices: intercept ($b_Intercept$), animal group (between-subject factor; $b_camera_newE.coli$), day (within-subject factor; $sd_Day_Intercept$), dispersion parameter (ϕ).

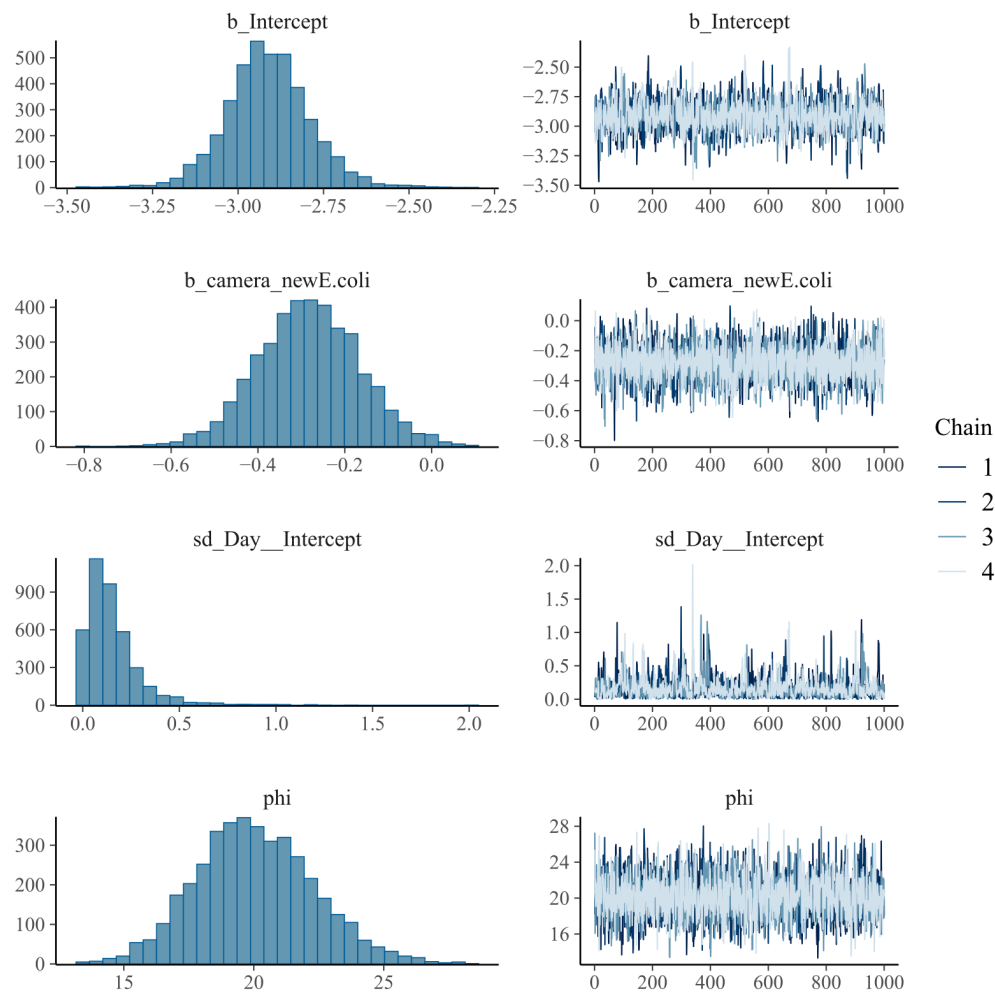


Figure S2. brms output for the change in body area, showing density plots for each factor (left) and caterpillar plots (right). Indices: intercept ($b_Intercept$), animal group (between-subject factor; $b_camera_newE.coli$), day (within-subject factor; $sd_Day_Intercept$), dispersion parameter (ϕ).

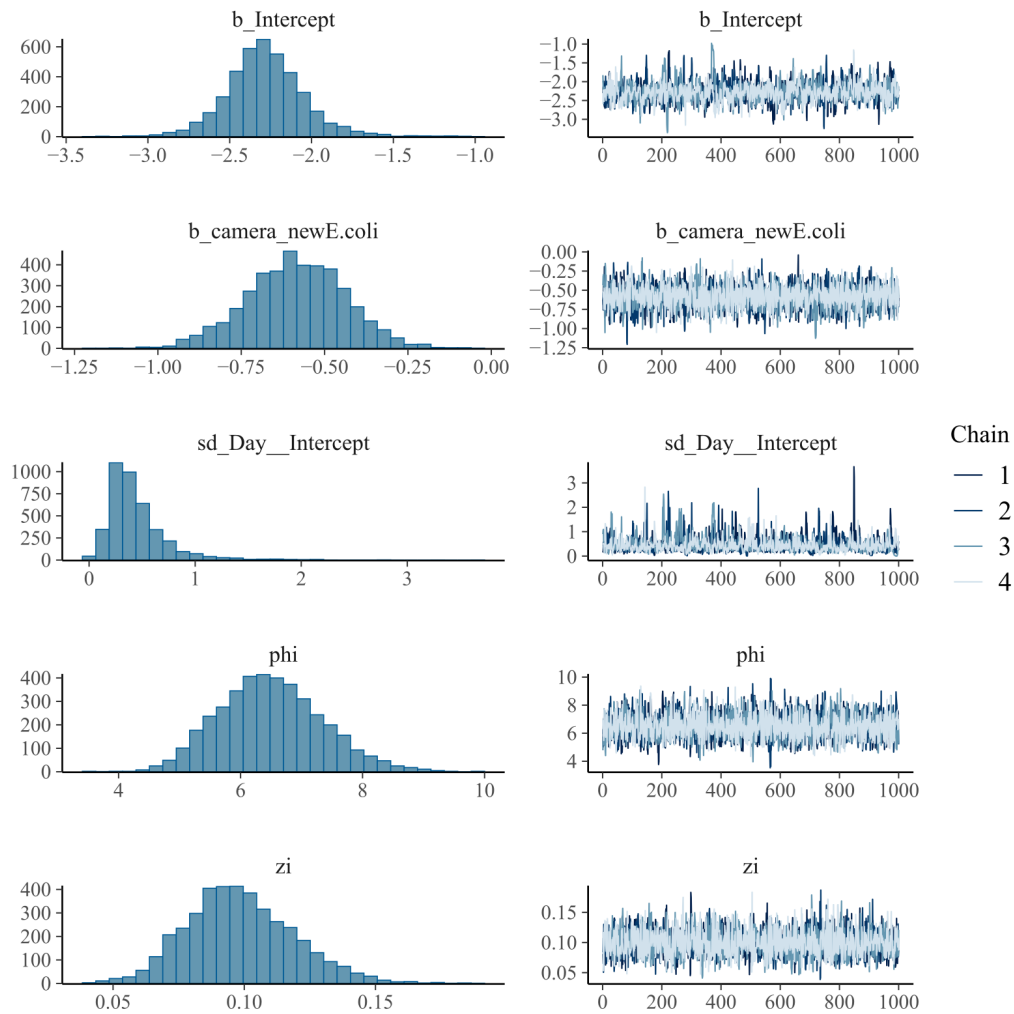


Figure S3. brms output for the time spent near the food source, showing density plots for each factor (left) and caterpillar plots (right). Indices: intercept ($b_Intercept$), animal group (between-subject factor; $b_camera_newE.coli$), day (within-subject factor; $sd_Day_Intercept$), dispersion parameter (ϕ) and zero-inflation parameter (zi).

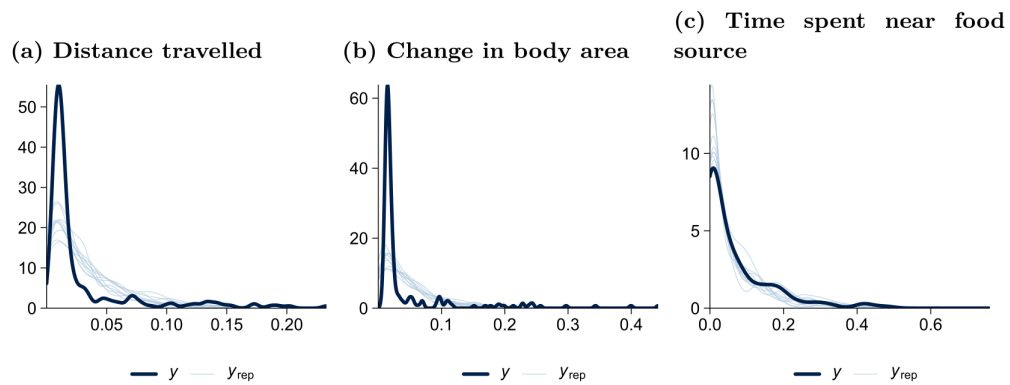


Figure S4. Posterior predictive checks for each model fit. y : observed outcome variable, y_{rep} : simulated datasets from the posterior predictive distribution.

A.2 Supplementary Tables

Table S1. Description of lesion score levels

Organ	Characteristic	Levels
Peritoneum	Inflammatory reaction	0: None 1: Local at the entrance 2: Around the ovary 3: In the omentum 4: All peritoneum
	Amount of exudate	0: None 1: Sparse 2: Some 3: Abundant
	Type of exudate	0: None 1: Aqueous 2: Lump of pus 3: Flakes 4: Confluent
	Transparency of peritoneum	0: Clear 1: Nuclear 2: Cloudy 3: Milky 4: Opaque
Liver	Inflammatory reaction	0: None 1: Fibrin covering < 25% 2: Fibrin covering > 25% 3: Fibrin covering all the liver
Airsac (left)	Transparency of airsac	0: Clear 1: Nuclear 2: Cloudy 3: Milky 4: Opaque
	Amount of exudate	0: None 1: Sparse 2: Some 3: Abundant
	Amount of exudate	0: None 1: Sparse 2: Some 3: Abundant
Lung (left)	Congestion	0: None 1: Sparse (< 25% of lung tissue) 2: Pronounced (> 25% of lung tissue)
	Lung lesions (oedema)	0: None 1: Slight oedema of the alveolar walls 2: Moderate oedematous thickening of alveolar walls with occasional alveoli containing coagulated oedema fluids 3: Extensive occurrence of alveolar and interstitial oedema
	Lung lesions (size)	0: No lesions 1: Small lesions between 1 cm ² and 2 cm ² 2: Moderate-size lesions between 3 cm ² and 4 cm ² 3: Extensive lesions with more than 4 cm ² and covering almost all the area of the lungs
	Granulomas	0: None 1: Few small white caseous/spherical nodules 2: Numerous small white caseous/spherical nodules 3: Many caseous lesions/nodules and green-grey mould due to sporulation
Spleen	Proliferation	0: None 1: Little proliferation 2: Distinct proliferation

Table S2. Mann-Whitney U tests to assess between-group differences in serum amyloid A levels, lesions scores, and number of colony-forming units.

A	B	U-val	<i>p</i> -value	Cohen's <i>d</i>
<i>Serum amyloid A (mg/L)</i>				
Control	<i>E. coli</i>	45.0	2.77e-05	-1.153
<i>Lesion score</i>				
Control	<i>E. coli</i>	39.5	1.064e-05	-2.121
<i>Colony-forming units (per gram of lung)</i>				
Control	<i>E. coli</i>	31.0	1.252e-06	-0.607

Table S3. *brms* summary tables for the behavioural features. Reference group: Control (uninfected). Significant differences (determined if the credible intervals exclude zero) with respect to the negative control are shown in bold. CI: credible interval (l: lower, u: upper). $\hat{R}$: Gelman-Rubin statistic for Markov chain Monte Carlo convergence; convergence is deemed suitable when $\hat{R}$ is close to 1. ESS: effective sample size.

	Estimate	Est.Error	l-95% CI	u-95% CI	$\hat{R}$	Bulk ESS	Tail ESS
<i>Distance travelled</i>							
Intercept	-3.32	0.116	-3.543	-3.079	1.003	2353.594	1771.796
<i>E. coli</i>	-0.286	0.112	-0.506	-0.069	1.002	3717.03	2639.991
<i>Rate of change in body area</i>							
Intercept	-2.912	0.125	-3.148	-2.658	1.001	1750.502	1543.872
<i>E. coli</i>	-0.283	0.117	-0.518	-0.049	1.001	3034.374	2758.279
<i>% Time near food source</i>							
Intercept	-2.281	0.245	-2.74	-1.761	1.002	1333.479	1306.324
<i>E. coli</i>	-0.585	0.148	-0.883	-0.302	1.001	3112.788	2167.955

Table S4. Comparison of different families for mixed-effects modelling using *brms*. ELPD: expected log pointwise predictive density. LOOCV: leave-one-out cross-validation. SE: standard error.

Family (link)	Δ ELPD		LOOCV ELPD	
	Estimate	SE	Estimate	SE
<i>Distance travelled</i>				
Beta (logit)			498.315	19.847
Gaussian (identity)	-153.134	8.685	345.181	19.389
<i>Rate of change in body area</i>				
Beta (logit)			418.771	22.354
Gaussian (identity)	-170.662	10.437	248.109	24.957
<i>% Time spent near food source</i>				
Zero-inflated Beta (logit)			292.026	31.327
Gaussian (identity)	-105.782	30.388	186.244	19.72